

A Short Review of Laplace Transform

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Notations

- $\mathbb{R}$ denotes the set of real numbers
- $\mathbb{C}$ denotes the set of complex numbers
- $\Re$ denotes the real part of a complex number, $\Im$ denotes its imaginary part
- $j := \sqrt{-1}$
- $F(s) = \mathcal{L}[f(t)]$ reads “ $F(s)$ is the Laplace transform of $f(t)$ ”

Definition of Laplace Transform

- Laplace transform $\mathcal{L}[\cdot]$ is a machine that takes a real function of time $t \in [0, \infty)$ and returns a complex function of the complex variable $s \in \mathbb{C}$
- $F(s) = \mathcal{L}[f(t)] := \int_0^\infty e^{-st} f(t) dt$
- Inverse Laplace transform $\mathcal{L}^{-1}[\cdot]$ undoes this operation, i.e., $f(t) = \mathcal{L}^{-1}[F(s)]$
- Domain of the function $F(\cdot)$ depends on the choice of the function $f(\cdot)$

Powers

- $\mathcal{L}[k] = \frac{k}{s}$ for constant $k \in \mathbb{R}$, check by direct integration, domain: $\Re s > 0$
- $\mathcal{L}[t^n] = \frac{n!}{s^{n+1}}$ for whole $n = 0, 1, 2, \dots$, check by induction, domain: $\Re s > 0$

Exponential, Sine, Cosine

- $\mathcal{L}[e^{zt}] = \frac{1}{s - z}$ for constant $z \in \mathbb{C}$, check by direct integration, domain: $\Re s > \Re z$
- Specialize the above for pure imaginary z , i.e., $z = j\omega$ with constant $\omega \in \mathbb{R}$
- ... to get two results in one shot:

$$\mathcal{L}[\cos(\omega t)] = \frac{s}{s^2 + \omega^2}, \quad \mathcal{L}[\sin(\omega t)] = \frac{\omega}{s^2 + \omega^2}, \quad \text{domain: } \Re s > 0$$

Examples

$$\mathcal{L}[e^{-5t}] = \frac{1}{s + 5}, \quad \mathcal{L}[2 \cos(3t) - 4 \sin(5t)] = 2 \cdot \frac{s^3 - 10s^2 + 25s - 90}{(s^2 + 9)(s^2 + 25)}$$

How to compute $\mathcal{L}^{-1} [\cdot]$

- Pattern recognition

Examples

$$\mathcal{L}^{-1} \left[\frac{7}{s+1} \right] = 7e^{-t}, \quad \mathcal{L}^{-1} \left[\frac{1}{s^2 + 36} \right] = \frac{1}{6} \sin(6t)$$

- If $F(s)$ is rational, then first do partial fraction expansion, then pattern recognition

Example

$$\begin{aligned} \mathcal{L}^{-1} \left[\frac{4s+5}{s^3(s+2)(s-1)^2} \right] &= \mathcal{L}^{-1} \left[\frac{69/8}{s} + \frac{23/4}{s^2} + \frac{5/2}{s^3} + \frac{1/24}{s+2} + \frac{-26/3}{s-1} + \frac{3}{(s-1)^2} \right] \\ &= \frac{69}{8} + \frac{23}{4}t + \frac{5}{4}t^2 + \frac{1}{24}e^{-2t} - \frac{26}{3}e^t + 3te^t \end{aligned}$$

Properties of Laplace Transform: Arithmetic

- Linearity

$$\mathcal{L}[af(t) + bg(t)] = aF(s) + bG(s) \text{ for } a, b \in \mathbb{C}$$

- Time scaling

$$\mathcal{L}[f(at)] = \frac{1}{|a|} F\left(\frac{s}{a}\right) \text{ for } a \in \mathbb{R} \setminus \{0\}$$

Properties of Laplace Transform: t Domain Calculus

- t domain derivative

$$\mathcal{L}[f^n(t)] = s^n F(s) - s^{n-1}f(0) - s^{n-2}\dot{f}(0) - \dots - f^{(n-1)}(0) \text{ for } n = 0, 1, 2, \dots$$

Examples

$$\mathcal{L}[\dot{f}] = sF(s) - f(0), \quad \mathcal{L}[\ddot{f}] = s^2F(s) - sf(0) - \dot{f}(0)$$

- t domain integral and convolution

$$\mathcal{L}[f * g] := \mathcal{L}\left[\int_0^t f(\tau)g(t-\tau)d\tau\right] = F(s)G(s)$$

Examples

$$\text{For } g(t) = 1 \text{ we get } \mathcal{L}\left[\int_0^t f(\tau)d\tau\right] = \frac{F(s)}{s}. \quad \text{Notice that } \mathcal{L}[f(t)g(t)] \neq F(s)G(s)$$

Properties of Laplace Transform: s Domain Calculus

- s domain derivative

$$\mathcal{L}[t^n f(t)] = (-1)^n F^{(n)}(s) \text{ for } n = 0, 1, 2, \dots$$

Examples

$$\mathcal{L}[tf(t)] = -F'(s), \quad \mathcal{L}[t^2 f(t)] = F''(s)$$

- s domain integral

$$\mathcal{L}\left[\frac{f(t)}{t}\right] = \int_s^\infty F(\sigma) d\sigma. \text{ The special case } s = 0 \text{ gives } \int_0^\infty \frac{f(t)}{t} dt = \int_0^\infty F(\sigma) d\sigma$$

Example

$$\mathcal{L}\left[\frac{\sin t}{t}\right] = \arctan \frac{1}{s}, \quad \int_0^\infty \frac{\sin t}{t} dt = \int_0^\infty \frac{d\sigma}{\sigma^2 + 1} = [\arctan \sigma]_{\sigma=0}^{\sigma=\infty} = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

Properties of Laplace Transform: Shift

- t domain shift

$$\mathcal{L}[f(t - \tau)] = e^{-s\tau} F(s) \text{ for } t > \tau$$

- s domain shift

$$\mathcal{L}[e^{at} f(t)] = F(s - a) \text{ for } a \in \mathbb{C}$$

Examples

$$\mathcal{L}^{-1}\left[\frac{2}{(s-3)^3}\right] = e^{3t}t^2, \quad \mathcal{L}^{-1}\left[\frac{1}{(s+5)^2+4}\right] = \frac{1}{2}e^{-5t}\sin(2t)$$

Initial Value Theorem (IVT)

- When applicable, IVT says: $\lim_{t \downarrow 0} f(t) = \lim_{s \uparrow \infty} sF(s)$
- Applicable when $\lim_{t \downarrow 0} f(t)$ exists and $f(t) = \mathcal{O}(e^{ct})$ for some $c \in \mathbb{R}$
- For rational $F(s)$, applicable if $F(s)$ is strictly proper
- Rational $F(s)$ is strictly proper if $\text{degree}(\text{denominator poly}) < \text{degree}(\text{numerator poly})$

Examples

IVT does not apply to $F(s) = \frac{s-1}{s+2}$ but applies to $F(s) = \frac{s-1}{s^2+s+2}$

Final Value Theorem (FVT)

- When applicable, FVT says: $\lim_{t \uparrow \infty} f(t) = \lim_{s \rightarrow 0} sF(s)$
- Applicable when $F(s)$ has no poles in closed right half plane (RHP) except at most one (i.e., non-repeated) pole at origin

Examples

FVT does not apply to $F(s) = \frac{2s + 5}{s^2(s + 3)}$ with poles 0, 0, -3 (repeated poles at origin)

FVT does not apply to $F(s) = \frac{s}{s^2 + 1}$ with poles $j, -j$ (both poles in closed RHP)

FVT does not apply to $F(s) = \frac{1}{s(s - 2)}$ with poles 0, 2 (even though 0 is non-repeated, 2 is in closed RHP)

FVT applies to $F(s) = \frac{1}{s(s + 2)}$ with poles 0, -2 (0 is non-repeated, -2 is NOT in closed RHP), so $\lim_{t \uparrow \infty} f(t) = \lim_{s \rightarrow 0} \frac{1}{s + 2} = \frac{1}{2}$. Check via $f(t) = \mathcal{L}^{-1}\left[\frac{1}{s(s+2)}\right] = \frac{1}{2}(1 - e^{-2t})$

Exercise

- Use the definition of Laplace transform to compute $\mathcal{L}^{-1}[1]$
- Compute $\mathcal{L}[7t^5 - 4\cos(3t) + 9e^{-t}]$
- Compute $\mathcal{L}[(t-2)^5]$ for $t > 2$
- Compute $\mathcal{L}^{-1}\left[\log\left(\frac{s-10}{s-20}\right)\right]$
- For $F(s) = \frac{s}{(s+5)(s^2+s+1)}$, use FVT, if possible, to compute $\lim_{t \uparrow \infty} f(t)$.
Compute the same without FVT